

An AI-Based Toolchain to Align Curricular Competencies with Internship Experiences

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Abstract—Higher education institutions increasingly need evidence-based insights to adapt curricula to rapidly changing professional practices. This paper presents an AI-based toolchain that analyzes student internship reports to identify sectoral contexts, work tasks, and competency profiles, enabling data-driven reflection on curriculum–practice alignment. The pipeline combines transformer-based text processing, topic modeling, and embedding-based similarity analysis to extract and compare competencies from 98 internship reports and 99 module descriptions across three interdisciplinary bachelor’s programs. Results reveal both alignment between curricular intentions and workplace activities as well as internship-dominant competencies not explicitly captured in curricula. Interpreted through a systems-theoretical lens, internship reports function as a medium of structural coupling between education and the labor market, extending institutional self-observation without replacing academic judgment. The study illustrates how AI-supported analysis can strengthen the reflexivity of higher education and support evidence-informed curriculum development.

Index Terms—educational data mining, large language models, curriculum alignment, systems theory, AI in higher education.

I. INTRODUCTION

Digitalization and the rapid diffusion of artificial intelligence (AI) are transforming both economic systems and professional practices at unprecedented speed. These transformations are accompanied by broader societal challenges—such as sustainability transitions and democratic resilience—that increasingly shape how work is organized and which competencies are valued. Higher education is thus confronted with a dual task: preparing students for professions that evolve continuously, while maintaining curricular stability and coherence. Yet curricular change is often delayed [1] because institutions lack systematic and timely mechanisms to observe developments in professional practice. As a result, curricular revisions typically rely on anecdotal feedback or periodic surveys rather than continuous evidence from the field.

Internships represent a particularly valuable but often underexploited source of such evidence (there is little empirical reporting of graduate skill transfer and inconsistencies in findings for learning transfer in general [2]). They capture a snapshot of the boundary between education and the labor market: students enact curricular competencies in real-world

contexts and document their experiences in reflective reports. These reports are routinely collected but rarely analyzed beyond administrative purposes. If processed systematically, they could provide an ongoing, data-driven medium through which the educational system observes signals from its economic environment. In systems-theoretical terms, they can serve as a medium of structural coupling (causality of structure (*Strukturkausalität*) [3]) between education and economy—allowing curricula to adjust based on observed practices rather than assumption [4].

However, unlocking this potential requires overcoming two persistent obstacles. First, internship reports are unstructured and diverse, making manual analysis resource-intensive and inconsistent. Second, privacy and data-protection concerns restrict large-scale manual processing. Advances in natural language processing (NLP) and large language models (LLMs) now make it feasible to automate such analyses while preserving confidentiality through local or hybrid AI deployment [5], [6]. By leveraging these technologies, it becomes possible to convert large volumes of textual reflections into structured, interpretable representations of professional practice.

This paper presents an AI-based toolchain designed to transform internship reports into an empirical interface between curricular design and labor market reality. The toolchain employs transformer-based text mining, topic modeling, and embedding-based similarity analysis to extract professional sectors, tasks, and associated competencies from students’ reports. These results are then compared with curricular learning outcomes, enabling educators to identify both alignments and gaps between institutional expectations and the competencies applied in practice.

The study focuses on three relatively new interdisciplinary bachelor’s programs—Informatics and Design, Digital Engineering, and Geodata Science—each of which integrates computer science with a distinct disciplinary domain. Informatics and Design combines programming, interaction design, and visual communication to prepare graduates for the creation of human-centered digital products and services. Digital Engineering merges informatics with engineering, equipping students to design, develop, and optimize intelligent and interconnected systems in industrial and production contexts.

Geodata Science integrates data science and geoinformatics, enabling students to analyze, model, and visualize spatial data to address environmental and societal challenges.

As these programs have only recently begun producing graduates, it is particularly relevant to observe the sectors in which students complete internships and the types of tasks they perform. These early professional trajectories offer valuable signals—rather than evaluative judgments—regarding how interdisciplinary curricular competencies manifest in practice and how emerging industries absorb digital and AI-related expertise. Analyzing internship reports makes it possible to detect such sectoral and task patterns systematically, thereby providing an empirical basis for institutional reflection.

Using 98 anonymized internship reports and 99 module descriptions, we demonstrate how AI-based analysis can uncover systematic patterns in students' professional experiences and their correspondence with curricular competencies.

The study is guided by the following research questions:

- RQ1** How can AI-based natural language processing transform internship reports into a medium that enables higher education institutions to systematically observe expectations and developments in professional practice?
- RQ2** How do patterns extracted from internship reports relate to the competencies articulated in curricular documents, and how can these relations inform institutional reflection?

By addressing these questions, this work contributes to the emerging field of educational data mining in higher education [7], [8] and to systems-theoretical approaches that conceptualize education as an adaptive, self-observing system [3], in which learning implies using what has already been learned as a basis for further learning [9, p. 92]. It also provides a practical demonstration of how AI can enhance institutional reflexivity [5] without replacing human judgment.

The remainder of the paper is structured as follows: Section II describes the related work. Section III states the methodological design and Section IV gives a detailed description of the processing pipeline; In Section V we describe our limited validation. Section VI presents results on sector classification, task extraction, and curricular mapping; Section VII discusses the results and points to methodological limitations of our approach and Section VIII concludes with a short summary of the article.

II. RELATED WORK

A. Curriculum–Practice Alignment and Internships

Aligning academic curricula with professional practice has long been a central concern in higher education [1], [10], particularly in applied and engineering-oriented programs. Prior research has addressed this challenge through curriculum mapping exercises, accreditation frameworks, employer surveys, alumni feedback, and structured evaluations of internship learning outcomes [1]. While these approaches generate valuable qualitative and normative insights, they are typically episodic, resource-intensive, and difficult to institutionalize as continuous, evidence-based observation practices.

Internships are commonly conceptualized as boundary experiences between education and work, supporting professional socialization, identity formation, and employability development [11]. Empirical studies of internship outcomes frequently rely on student self-reports, supervisor evaluations, or predefined competency frameworks. Recent qualitative and mixed-method research has highlighted both the prevalence and the limitations of these approaches, noting their tendency to privilege explicit, self-perceived learning and standardized indicators while overlooking tacit, situated, and emergent learning processes [12]–[14]. Although internship reports are routinely collected as reflective artefacts, they are rarely treated as a longitudinal or cohort-level data source for institutional learning and curriculum development [13], [14].

As a result, much of the existing literature emphasizes retrospective evaluation of curriculum–practice alignment rather than enabling ongoing observation of how professional practice manifests in student experiences across cohorts. This gap motivates analytical approaches that complement established qualitative methods by supporting scalable, data-informed, and longitudinal curriculum reflection.

B. Educational Data Mining and NLP in Curriculum Analysis

Educational data mining has long employed text mining techniques to analyse unstructured educational data, including discussion forums and open-ended student responses [7]. More recently, both educational data mining and learning analytics have increasingly adopted natural language processing (NLP) methods to enable scalable analysis of such data, including discourse analysis, sentiment analysis, and topic modelling [8], [15].

Beyond learner-generated content, NLP has also been applied to large-scale analysis of unstructured curricular documents such as course syllabi and learning outcome descriptions to infer educational content and skill coverage [16]. A growing body of work employs transformer-based models, including BERT-based semantic retrieval [17], [18] and large language model (LLM) prompting [19], to analyse textbooks, curriculum guidelines, and accreditation documents in support of curriculum mapping and quality assurance [20]. This line of research has increasingly extended toward education–occupation alignment, with NLP-based frameworks that jointly analyse curricular documents and labour-market texts using skill extraction, embedding-based representations, and semantic similarity measures, often aligned with standardized taxonomies such as ESCO [21], [22].

A wide range of NLP techniques—including topic modelling, keyword extraction, text classification, and semantic similarity analysis—have been applied to reflective essays, feedback comments, and course evaluations to support assessment, prediction, and learning outcome analysis [23], [24]. Recent studies indicate that deep learning models such as BERT generally outperform traditional machine-learning and lexicon-based approaches on many of these tasks [18]. While LLM-based approaches do not yet consistently surpass task-specific supervised models in predictive accuracy, recent

TABLE I
EXAMPLES OF EXTRACTED LEARNING GOALS FROM MODULE DESCRIPTIONS TRANSLATED INTO ENGLISH.

evidence suggests that they are particularly effective at extracting implicit, context-dependent, and syntactically complex skills when combined with embedding-based normalization and standardization mechanisms [25], [26].

C. Positioning of the Present Work

In contrast to prior work that focuses on automated assessment, benchmarking, or validation of curriculum–practice alignment, this paper introduces an AI-based toolchain designed for exploratory institutional observation. The goal is not to evaluate student performance or curricular adequacy, but to generate structured, interpretable representations of internship experiences that can inform reflective curriculum development at the program level.

Conceptually, the approach is grounded in systems-theoretical perspectives that conceptualize education as a self-observing system capable of reflecting on its relations with its environment [3], [9]. From this perspective, internship reports function as a medium of structural coupling between higher education and professional practice.

III. METHODOLOGY

This study follows a mixed computational–analytical research design aimed at transforming unstructured internship reports into structured representations of professional practice and comparing them with curricular competence profiles. The overall objective is not evaluative benchmarking but diagnostic pattern detection that supports institutional reflection on curriculum–practice alignment.

We draw on the internship reports of 98 students, which document their mandatory, semester-long paid internships in industry. These reports are conceptualized as empirical traces of students’ engagement with professional environments. Through systematic text analysis, these traces are treated as indicators of sectoral contexts, professional tasks, and applied competencies. The methodological logic combines (1) unsupervised pattern discovery to obtain an interpretation-agnostic overview of the data, with (2) targeted LLM-based extraction to identify interpretable categories such as sectors and tasks. Finally, embedding-based similarity analysis is used to compare internship-derived competencies with intended curricular learning outcomes.

The pipeline is designed to generate structured representations of internship experiences that support institutional reflection rather than automated evaluation. Accordingly, the outputs of the toolchain are not treated as findings about reality, but as second-order observations that require interpretation and contextualization by educators. They function as signals that may prompt further inquiry, discussion, or qualitative review,

rather than as definitive assessments of curricular adequacy or competence coverage.

IV. IMPLEMENTATION

The proposed AI-based toolchain follows a modular, sequential architecture designed to transform unstructured internship reports into structured representations that support curricular reflection. The architecture consists of six stages: (A) data ingestion, (B) preprocessing and anonymization, (C) unsupervised topic modelling and embeddings to establish a baseline semantic structure, (D) LLM-based sector extraction and classification, (E) extraction of student-executed tasks and their cognitive characterization, and (F) curricular mapping.

The architecture combines local processing components with LLM-based modules, enabling flexible deployment depending on institutional constraints.

A. Data Ingestion

The empirical basis of the study comprises two text corpora:

- **Internship Reports:** The dataset comprises 98 anonymized internship reports submitted between 2021 and 2024 by undergraduate students enrolled in three interdisciplinary bachelor’s programs at Hochschule München University of Applied Sciences: Informatics and Design ($n = 68$), Digital Engineering ($n = 25$), and Geodata Science ($n = 5$). Reports were written in German, range from 10–30 pages ($\mu = 3138.93$ words, $\sigma = 1076.23$), and document tasks, learning experiences, and reflections during a mandatory internship semester.
- **Module Descriptions:** To represent the curricular perspective, 99 module descriptions in reStructuredText format were collected from the same programs. These documents contain learning objectives and intended learning outcomes.

B. Preprocessing and Anonymization

Internship reports were converted to plain text using a hybrid PDF extraction workflow with OCR fallback (Tesseract v5). A custom Python pipeline based on spaCy v3.7 [27] (using a pre-trained language-specific processing pipeline called `de_core_news_lg`¹) performed sentence segmentation, error correction, and removal of non-informative fragments (e. g., duplicates or very short sentences). After preprocessing, 16,091 sentences (72% of the original corpus) remained, with an average length of $\mu = 15.83$ words ($\sigma = 9.00$).

¹It uses a general-purpose model for the German language trained on news-style text.

Privacy preservation was a core requirement. Anonymization combined rule-based and machine-learning-based named entity recognition, followed by secondary validation using a locally hosted LLM (DeepSeek-r1:14b [28] via Ollama [29]) to detect residual personally identifiable information. Personal names and sensitive identifiers were removed. A manual check of 10% of the reports confirmed compliance.

To enable topic modelling, internship reports were segmented into overlapping *sliding windows* of seven sentences with a three-sentence overlap, yielding 5,261 pseudo-documents ($\mu = 111.97$ words, $\sigma = 33.81$). Module descriptions were reduced to competence-related sections, resulting in 1,798 sentences ($\mu = 11.04$ words, $\sigma = 8.13$). Example learning outcomes are shown in Table I.

Two configurations were used for subsequent analyses: (1) a fully local pipeline using open-source models, and (2) a hybrid pipeline incorporating commercial LLMs.

C. Topic Modelling and Embeddings

Topic modeling was applied to identify recurring themes in both internship reports and module descriptions. We employed BERTopic [30], combining transformer-based embeddings with class-based TF-IDF weighting [31]. All text segments were embedded using the pre-trained paraphrase-multilingual-mpnet-base-v2 SentenceTransformer [32], mapping German text into a 768-dimensional vector space. Dimensionality reduction was performed using UMAP [33], followed by clustering with HDBSCAN. Representative keywords were extracted using class-based TF-IDF, and topic labels were refined using GPT-4o-mini [34].

Read the following internship report written in German and extract the industries in which the student worked.

Requirements:

- Identify specific industries (not overly general terms such as “Technology” or “Services”).
- List only industries that can be clearly inferred from the text.
- Provide a maximum of 3 relevant industries.
- Output format: a JSON list of short phrases (max. 2 words per industry, in German).
- No additional explanations, no continuous text, and do not include “json.”

Text: [Internship report]

Fig. 1. **Prompt 1:** Sector extraction from internship reports.

D. Sector Distribution Estimation

To identify the economic sectors represented in the internship experiences, we applied a three-stage procedure.

a) *GPT-based Sector Extraction:* Each report was processed with GPT-4o-mini using a structured prompt (Fig. 1) requesting the most probable industry sectors (in German). For robustness, five independent runs were collected per report (temperature = 0.1), resulting in 336 initial sector terms (109 for Digital Engineering, 235 for Informatics and Design, 27 for Geodata Science). Using GPT-4o-mini for sector and later for task extraction is justified by recent findings showing that LLMs can effectively perform inductive coding tasks traditionally done by human experts [35].

b) *Clustering of Sector Terms:* To consolidate semantically similar sector labels, the extracted terms were clustered using the same BERTopic pipeline described above. Labels were reviewed and refined to improve consistency across programs.

c) *Zero-shot Sector Classification:* A multilingual entailment pre-trained model, MoritzLaurer/mDeBERTa-v3-base-mnli-xnli, was used to classify each sliding window into one of the clustered sector labels. The resulting scores were aggregated to derive sector distributions per report. Following established practice, these scores are treated as indicators rather than probabilities.

Read the following excerpt from an internship report (in German) and identify all concrete tasks/activities that the student actually carried out themselves. Observations, goals, mere descriptions of learning, theoretical content, or activities performed by other people must not be listed.

Assign each task a cognitive demand level according to Webb’s Depth of Knowledge (DOK) (Level 1–4). Use the following criteria:

- DOK1: Reproduction or use of known, routine steps with no room for decision-making
- DOK2: Procedure involving selection, adaptation, or comparison; multiple possible approaches
- DOK3: Justified planning, analyzing, developing solution strategies for non-routine problems
- DOK4: Complex, multi-step, longer-term problem-solving over time with transfer or investigation

Provide the result as pure JSON, without code block and without any additional explanations. Format:

Text: [Internship report]

Fig. 2. **Prompt 2:** Task extraction and ranking from internship reports.

E. Task Extraction and Clustering

To extract the tasks students performed during their internships, we used GPT-4o-mini to identify concrete student-executed activities from report segments. Tasks merely observed, planned, or described theoretically were excluded. For each task, the model also estimated cognitive demand using Webb’s Depth of Knowledge (DOK) [36], [37], distinguishing routine procedural activities (Level 1) from complex problem-solving or extended reasoning (Level 4), see Fig. 2. Extracted task statements were then embedded and clustered with BERTopic to identify recurring professional activity patterns.

F. Curricular Mapping

To analyze alignment between curricular intentions and internship experiences, we compared embeddings of extracted internship tasks with embeddings of module description sentences. A combined BERTopic model was fitted to both corpora, enabling topics to be categorized as: (1) curriculum-only, (2) internship-only, or (3) shared.

Topics dominated by internship tasks but weakly represented in module descriptions were flagged as *potential areas of asymmetric representation*, i.e., areas where students engaged in professional activities not explicitly reflected in the curriculum. These preliminary findings serve as starting points for curricular reflection rather than definitive judgments of absence, recognizing that some learning outcomes are implicit

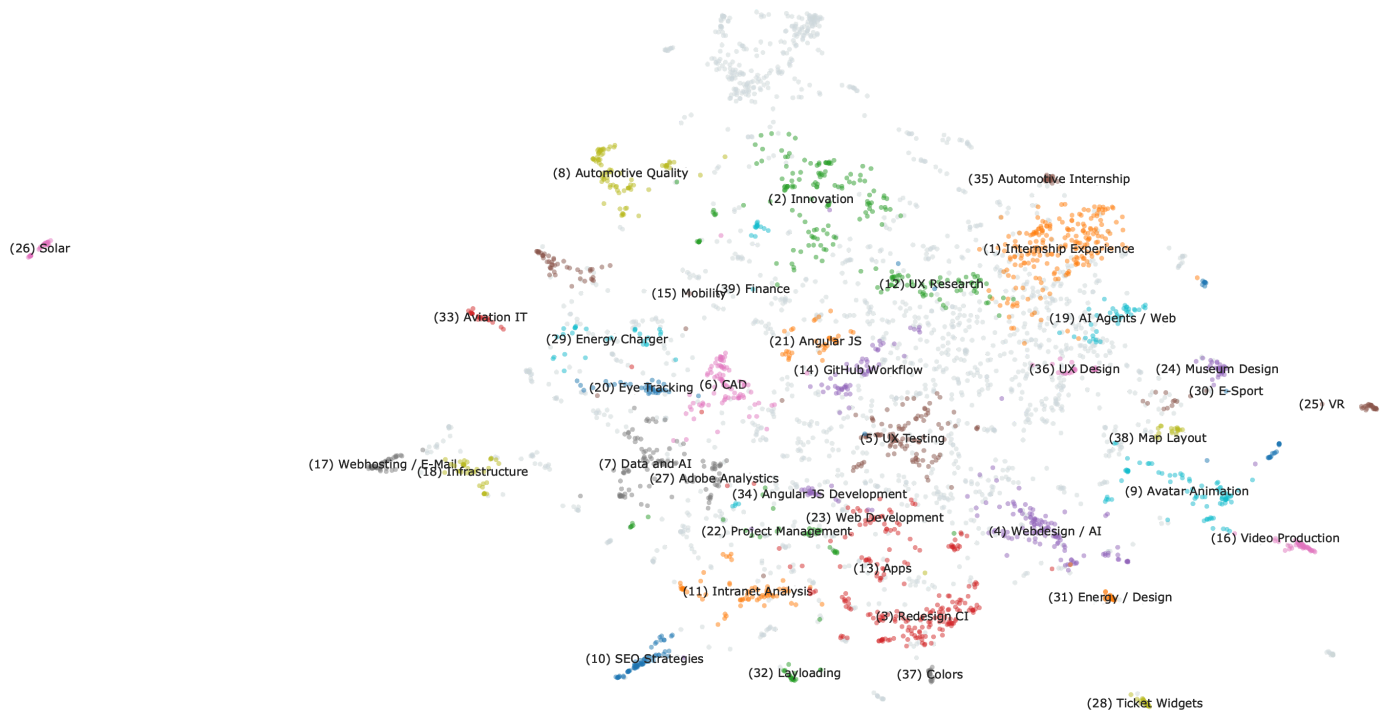


Fig. 3. Two-dimensional (projected) embedding of sliding windows from internship reports, used to explore recurring semantic themes prior to any LLM-based labeling. Distances reflect semantic similarity in the embedding space, not quantitative differences or topic boundaries. The visualization supports exploratory corpus inspection rather than categorical classification. Online version: https://bzoennchen.github.io/internship-analysis/fig_win_id.html.

in interdisciplinary program design and others do not have to be learned at university.

V. VALIDATION

Because internship reports do not have an objective ground truth and curricular interpretation is inherently normative, traditional accuracy-based validation (e.g., precision/recall against gold labels) is not appropriate. Therefore, validation in this study focuses on plausibility, robustness, and interpretability rather than predictive accuracy. Several measures were implemented to support credibility. First, anonymization quality was verified through manual inspection of a random subset of reports. Second, LLM-based extraction steps (sector and task identification) were run multiple times with low temperature settings to reduce stochastic variance. Third, unsupervised topic modelling was conducted prior to LLM-based labelling to provide an interpretation-agnostic baseline of corpus structure.

Classification outputs were explicitly treated as second-order observations intended to inform human judgment rather than ground-truth labels. Human spot checks of extracted sectors, tasks, and topic labels suggested generally coherent results, but systematic expert validation was beyond the scope of this study. Limitations arising from reliance on student self-reporting, concise module descriptions, and the use of commercial LLMs are acknowledged and discussed in Section VIII

VI. RESULTS

This section presents the findings of the AI-based analysis across the three bachelor's programs. We report on (A) topic overview, (B) estimated sector distribution, (C) extracted tasks and their estimated cognitive demand, and (D) correspondence between internship activities and curricular learning outcomes.

A. Identified Topics

Before applying LLM-based extraction methods, topic modelling was conducted on the sliding windows of internship reports to obtain an initial, LLM-agnostic overview of the semantic structure of the corpus, see Fig. 3. Using BERTopic with SentenceTransformer embeddings, this unsupervised analysis produced coherent clusters that reflect recurring themes across the reports. This step provides a baseline representation of the corpus structure before introducing model-generated labels.

The resulting topic landscape already provided a meaningful representation of the content without requiring any LLM interpretation. Clear thematic patterns emerged, including typical tasks (e.g. SEO, animation, UI/UX-design), sector-related contexts (IT, event maps), and technologies (e.g. solar, eyetracking) or tools (e.g. Angular, Adobe) used during internships. While Informatics and Design reports exhibit a broad dispersion of topics leaning towards web-technologies and design Digital Engineering reports cluster more around software development, IoT and data processing. Both reveal the strong automotive industry in Munich.

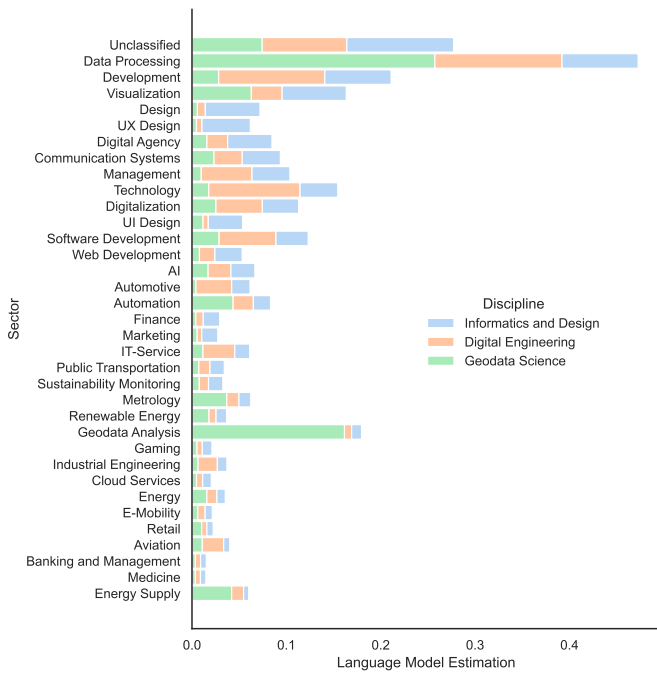


Fig. 4. Aggregated sector classification scores across all internship report segments. Scores indicate the relative prominence of sector-related contexts in student reports, not the proportion of internships per industry or verified employment statistics. Results should be interpreted as contextual signals rather than formal sector assignments.

These clusters offer a comparatively interpretation-agnostic starting point for understanding student experiences (while acknowledging that biases [38] inherent in embeddings and topic modeling remain), allowing researchers and educators to explore the data before introducing LLM-based labeling or human interpretation.

The visualization of the topic models in an interactive 2-dimensional embedding space enables further exploratory analysis. Users can inspect clusters, navigate between topics, and review representative text excerpts that contributed to each topic. Optional interactive visualizations of the topic models are provided as supplementary material [39]. These visualizations present the same results shown in static form in this paper and are not required for interpreting the findings.

B. Estimated Sector Distributions

Across the 98 internship reports, the toolchain identified a diverse set of industry sectors in which students gained professional experience. The combination of LLM-based sector extraction, topic clustering, and zero-shot classification enabled a structured estimation of sector representation without requiring manual coding of full reports. However, sector labels should be interpreted as contextual frames rather than precise economic classifications. Some labels (e.g., software development, data processing) reflect dominant activity contexts rather than formally distinct industries.

For Digital Engineering, 13 sectors emerged, including automotive, communication systems, industrial engineering, e-

mobility, energy supply, software development, data processing and AI, and automation. This reflects a strong orientation toward engineering domains in which digitalization plays an increasingly central role.

For Geodata Science, three dominant sectors were identified: geodata analysis, sustainability monitoring & energy, and software development. Although the sample size is small, findings indicate that students apply geospatial expertise in environmentally and socially relevant contexts. The concentration of renewable energy and sustainability-related sectors, despite the small sample size, illustrates how even limited data can surface program-specific contextual signals.

Informatics and Design exhibited the broadest sector distribution, with students engaging in software and web development, digital agencies, design and UX/UI, automotive, public transportation, renewable energy, marketing, IT services, and data visualization. This variety reflects both the interdisciplinary profile of the program and the versatility of digital design skills.

Unsurprisingly, terms related to software development and data processing were prominent across all programs. While these may not always reflect distinct “sectors” in a strict economic classification sense, their prominence indicates the centrality of digital transformation activities in students’ internships. Fig. 4 visualizes the sector scores aggregated by discipline.

Overall, the sector analysis demonstrates that students are exposed to a wide range of professional environments, with program-specific patterns consistent with expected disciplinary orientations.

C. Tasks and Estimated Cognitive Demand

Task extraction yielded a detailed overview of the types of work students engaged in during their internships, see Fig. 5. Using GPT-4o-mini, we extracted concrete student-executed activities and classified them according to Webb’s Depth of Knowledge (DOK). DOK levels are used here as a descriptive heuristic to characterize the cognitive demands of reported tasks, not as an assessment of students’ competence or learning outcomes. Tasks ranged from routine coding and documentation to complex problem-solving and design processes.

Across all programs, DOK2 and DOK3 tasks were most common, reflecting typical entry-level responsibilities such as implementing predefined features, adapting existing solutions, conducting testing, or following established design and development procedures. These tasks involve procedural decision-making and analytical thinking rather than simple reproduction. This predominance aligns with the typical role of interns as contributors within predefined workflows, while the presence of DOK4 tasks in a subset of cases points to opportunities for extended problem-solving under specific conditions. DOK1 and DOK4 tasks were less frequent, but still present in all programs, see Fig. 6.

In Informatics and Design, tasks often combined technical implementation with design-oriented activities (e.g., user research, wireframing, interaction design). Digital Engineering

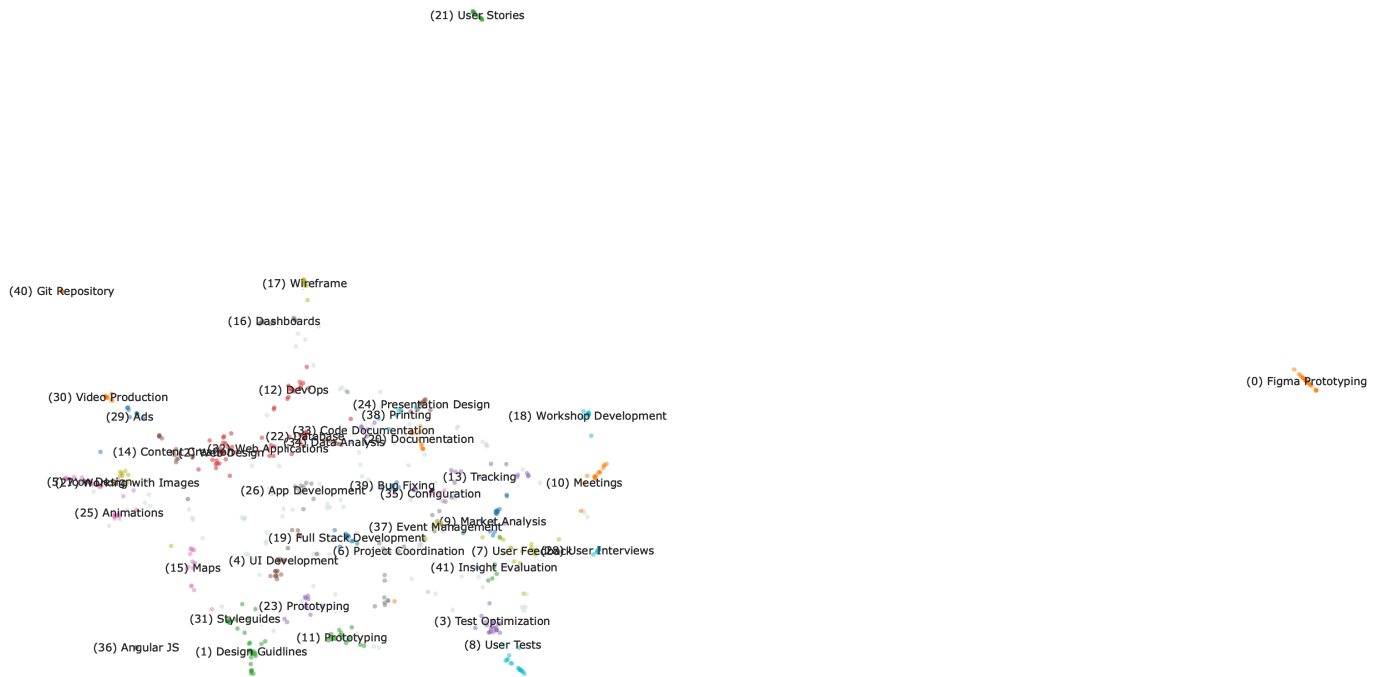


Fig. 5. Two-dimensional (projected) embedding of tasks extracted from internship reports (Informatics and Design). The visualization illustrates the semantic diversity and recurrence of reported activities (41 topics). Proximity reflects similarity in task descriptions, not task difficulty, importance, or curricular priority. Online version: https://bzoennchen.github.io/internship-analysis/fig_tasks_id.html.

students more frequently engaged in software development in technical industrial contexts, including embedded systems and automation. Geodata Science students conducted analytical tasks involving spatial data processing and visualization, with some instances of DOK3-level investigative work.

The distribution of DOK levels suggests that while internships provide valuable hands-on experience, most tasks remain within procedural or moderately complex cognitive demands. Instances of higher-order knowledge application do occur, indicating opportunities for independent problem-solving and creative design, though typically for a subset of students.

D. Curricular Mapping

The embedding-based comparison of internship tasks with module learning outcomes enabled an initial assessment of areas of asymmetric representation. Joint topic modelling across internship tasks and module description sentences revealed three categories of topics:

- (1) **Shared Topics:** Areas well represented in both internships and curriculum (e. g., software development, interface design, data processing). These indicate alignment between intended learning and workplace practice.
- (2) **Curriculum-Dominant Topics:** Competencies emphasized in the curriculum but encountered less frequently in internships (e. g., theoretical foundations, scientific methods). This reflects the different logics of educational vs. professional practice.
- (3) **Internship-Dominant Topics:** Topics that appeared frequently in internship activities but were weakly represented in module descriptions. These suggest potential

blind spots or areas worth reviewing for curricular enhancement.

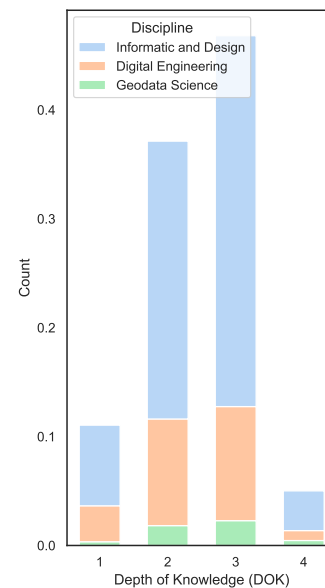


Fig. 6. Distribution of Webb's Depth of Knowledge (DOK) levels as estimated for extracted internship tasks. DOK levels (1–4) are used as a descriptive heuristic for characterizing reported task demands, not as an assessment of student competence or learning outcomes.

For Informatics and Design, six out of 29 topics fell into the third category and appeared to be “internship-dominant”, see Fig. 7. Examples included Figma prototyping, web design

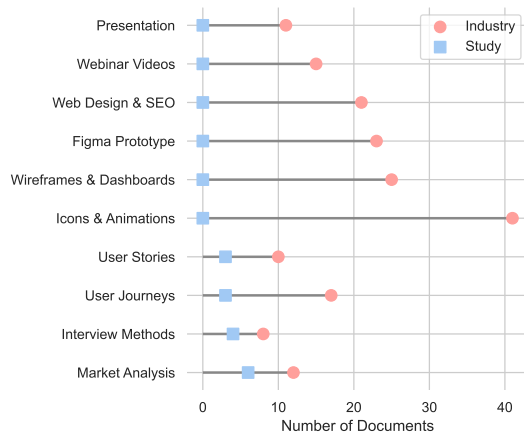


Fig. 7. Topics with comparatively low representation in module descriptions (< 10 data points) for the Informatics and Design program. These topics highlight areas where professional practices are articulated more explicitly in internship reports than in curricular documentation. They are not indicators of curricular absence or deficiency.

and SEO, user journeys, and presentation or webinar creation. While some of these competencies may be implicitly present in project-based coursework, they were not prominently reflected in module descriptions. No uncovered (internship-dominant) topics were identified for Digital Engineering or Geodata Science. This may reflect the rapid evolution of design tools and practices, which are often integrated implicitly into project-based courses rather than specified as standalone learning outcomes.

Two interpretive considerations are important. Internship-dominant topics do not necessarily indicate missing curricular content. Rather, they highlight areas where professional tools or practices are more explicitly named in internship reports than in curricular documentation. Module descriptions are often concise and do not capture all competencies taught in practice. Thus, the absence of explicit references cannot be taken as evidence of curricular omission. Second, interdisciplinary competencies—particularly soft skills and applied design tools—tend to surface implicitly rather than as dedicated learning outcomes. Therefore, human interpretation and reconsideration is needed.

Nevertheless, the approach highlights areas where curricular designers may wish to examine whether currently implicit competencies should be made more explicit, strengthened, or differently scaffolded. The method functions not as an evaluative audit but as an evidence-based stimulus for curricular reflection.

VII. DISCUSSION

A. Interpretation of Results and Response to RQ1

RQ1 asked how AI-based NLP can transform internship reports into a medium for systematic institutional observation of professional practice. The results demonstrate the feasibility of such a transformation at three levels. First, at an operational level, the toolchain enables the processing of large numbers

of unstructured internship reports without manual coding, using a combination of local preprocessing and scalable AI-based analysis. In this study, 98 multi-page reports were converted into structured representations of sectors, tasks, and competency-related patterns, illustrating the practical viability of the approach for routine institutional use.

Second, at an analytical level, the combination of unsupervised topic modeling and targeted LLM-based extraction produced coherent and program-specific patterns that align with expected disciplinary profiles. The stability of these patterns across multiple model runs and analytical steps suggests that the toolchain generates robust signals rather than idiosyncratic outputs.

Third, at an institutional level, the outputs are interpretable at the topic and pattern level and can be related to existing curricular documents. Importantly, these outputs are not treated as objective measurements but as second-order observations that require interpretation by educators. In this sense, the toolchain does not replace curricular judgment but extends the institution's capacity to observe recurring features of professional practice reflected in student reports.

Taken together, these findings indicate that AI-based NLP can feasibly transform internship reports into a structured observational medium, provided the outputs are used for reflective and dialogical purposes rather than automated evaluation.

B. Interpretation of Curriculum–Internship Relations and Response to RQ2

RQ2 addressed how patterns extracted from internship reports relate to competencies articulated in curricular documents. The results show that embedding-based comparison and joint topic modelling can surface areas of overlap as well as asymmetric representations between internship experiences and formal curricula.

Shared topics indicate areas where curricular learning outcomes and reported internship activities converge, suggesting continuity between intended and enacted competencies. Internship-dominant topics, by contrast, do not indicate missing or deficient curricula. Rather, they highlight professional tools, practices, or contextualized activities that are more explicitly articulated in student reports than in module descriptions, which often condense multiple learning outcomes into brief formulations. These asymmetries therefore function as prompts for curricular reflection rather than as evidence of misalignment.

Differences observed across study programs further illustrate the contextual nature of these patterns. Programs with broader or more practice-oriented profiles exhibit a wider range of internship-dominant topics, whereas more technically focused programs show tighter correspondence between curricular and internship-derived patterns. For programs with smaller cohort sizes, such as Geodata Science, the resulting patterns should be interpreted as illustrative rather than stable representations, yet they nonetheless demonstrate the sensitivity of the approach to program-specific contexts.

Overall, the analysis supports the use of AI-based representations as a means to explore relations between curriculum and practice within a bounded institutional setting. It does not support general claims about curricular adequacy or transferability across institutions, but rather demonstrates how such relations can be made observable for local interpretive use.

C. Methodological Limitations and Interpretive Boundaries

The interpretation of these results is subject to several methodological boundaries that are intrinsic to the chosen research design. First, sector labels, task descriptions, and Depth of Knowledge classifications are generated using LLMs and are not evaluated against an external ground truth. Given the absence of objective reference standards for internship experiences, the analysis prioritizes stability, coherence, and interpretability over accuracy-based validation. As a result, the outputs should be understood as exploratory signals rather than definitive classifications.

Second, the mapping between internship activities and curricular competencies is influenced by the granularity of curriculum documentation. Module descriptions often aggregate multiple learning outcomes into single sentences, which may lead to underrepresentation of certain competencies in the comparison step. This limitation constrains interpretation of curriculum–internship asymmetries and underscores the importance of reading internship-dominant topics as documentation effects rather than curricular gaps.

Third, sector-related patterns are subject to conceptual ambiguity, as terms such as “software development” or “data processing” can describe activities rather than formally defined industries. While clustering and zero-shot classification help mitigate this ambiguity, refined taxonomies or standardized industry classifications could further enhance conceptual clarity.

Fourth, the analysis relies on student self-reporting. Internship reports vary in narrative depth, reflective style, and specificity, meaning that extracted patterns reflect reported rather than directly observed professional practice. This introduces variability that shapes which activities become visible through the analysis and reinforces the need for cautious interpretation.

Finally, the use of commercial LLMs introduces limitations related to transparency, long-term reproducibility, and model drift. To mitigate these effects, the toolchain emphasizes architectural transparency and pattern-level stability rather than exact reproducibility of outputs. Future work may explore open or locally deployable models to increase methodological control.

D. Implications for Curriculum Development and Educational Data Mining

Despite these limitations, the findings contribute to ongoing work in educational data mining and learning analytics by demonstrating how AI-based text analysis can support institutional reflection rather than automated decision-making. By conceptualizing internship reports as a medium for organizational self-observation, the approach aligns with systems-

theoretical perspectives on education as an adaptive, self-observing system.

For curriculum development, the toolchain offers a scalable means to surface recurring patterns in professional practice that can inform dialogical curriculum review processes. Rather than replacing qualitative judgment, it provides an additional analytic lens that can help educators identify areas for discussion, clarification, or further inquiry.

From a systems-theoretical perspective, internship reports function as a medium of structural coupling between higher education and the labor market. They enable the educational system to register recurring features of professional practice without collapsing curricular logic into immediate workplace demands. Importantly, systems-theoretical accounts of education emphasize that a smooth or frictionless transition from university to work can neither be fully ensured nor is normatively desired [9]. Educational programs must therefore retain the capacity to cultivate forms of knowledge and competence that are not reducible to current occupational practices.

VIII. CONCLUSION

Internship reports have traditionally served as reflective assignments for students and evaluative artefacts for academic staff, but not as a data source for institutional learning. This paper demonstrated how AI-based text analysis can transform such reports into structured signals that support institutional self-observation and evidence-informed curriculum reflection.

By combining topic modeling, embedding-based similarity analysis, and targeted large language model extraction, the proposed toolchain enables scalable analysis of internship experiences without manual reading of all reports. Rather than producing evaluative judgments, the approach provides interpretable patterns that make relations between curricular intentions and professional practice observable.

Interpreted through a systems-theoretical lens, internship reports function as a medium of structural coupling between education and the labor market. They allow the educational system to register recurring features of professional practice while translating them into its own communicative code. Importantly, this does not imply that curricula should mirror workplace practices directly; universities retain responsibility for teaching competencies that extend beyond immediate job requirements.

Future work will focus on improving reproducibility and methodological control through alternative language models, refining specific pipeline components via partial manual annotation, and expanding the empirical basis across cohorts and institutions. Overall, the study illustrates how AI-supported analysis can enhance institutional reflexivity and support ongoing, dialogical curriculum development.

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